

Tomasz Frelek

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EDUCATION

University of California San Diego

GPA: 4.00

M.S. Computer Science, Research Distinction

San Diego, CA

2025-Present

The Ohio State University

GPA: 3.88

B.S. Computer Science Engineering, A.I. Specialization

B.S. Mathematics, Theoretical Track

Columbus, OH

2021-2025

INDUSTRY EXPERIENCE

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Radar Perception Engineer

Dayton, OH

Summer 2025

- Developed BEV transformer models for 3D object detection and semantic mapping, enhancing the perception system's MOTA score by 15%.
- Engineered a modular model evaluation package for object detection models. The tool assessed performance using MOTA metrics and custom, definable parameters, which accelerated feedback cycles 2X.
- Engineered a synthetic data pipeline using CARLA and DSpace simulators to generate realistic radar data, generated 100,000 frames of synthetic data, improving model performance in complex and hard-to-reproduce scenarios.
- Contributed to project preparations for the Consumer Electronics Show (CES) by developing tools for data synchronization and calibration, including: creating camera calibration matrices, implementing a system to align global and local positioning data for accurate data merging, and synchronizing GPS and local street data to build comprehensive route representations.

JPMorgan Chase

Software Engineer

Columbus, OH

Summer 2023/2024

- Used Python and industry standard frameworks like Flask and Unittest to create an internal interface for Amazon S3 that streamlined access to data, increased data access speed by 50%, and facilitated faster software development firm-wide across hundreds of development teams
- Executed software development tasks using Java, SpringBoot, and React to develop a comprehensive health check application for a data pipeline service, facilitating the transfer of petabytes of data for dozens of development and regulatory teams.
- Leveraged CI/CD frameworks like, Jenkins and Maven, to streamline application development and deployment.
- Utilized Agile development principles and Git for version control to collaborate with dozens of inter-departmental development teams on the development of a data pipeline interface utilized by 500+ of developers.

RESEARCH EXPERIENCE

Time Series Classification and Forecasting

Research Assistant — Advisor: Prof. John Paparrizos

Columbus, OH

Jan 2024 – May 2025

- Conducting research focused on implementing time series classification and forecasting algorithms and experimentally analyzing their efficacy.
- Building a comprehensive library of 60+ time series classification algorithms, implementing a unified interface for seamless integration into academic and industrial applications.
- Reading and analyzing novel industry papers to extract insights from cutting-edge methods and theories. Applying this knowledge by implementing algorithms and methodologies outlined in literature
- Utilizing Pytorch to develop, implement, and train deep-learning algorithms for time series classification.

Computer Vision for Ecological and Autonomous System Applications

Research Assistant — Advisor: Prof. Wei-Lun Chao

Columbus, OH

Aug 2024 – Aug 2025

- Pioneering Imageomics applications, integrating machine learning and computer vision for biological data analysis. Developing models for tasks such as species classification, trait labeling, and image segmentation.
- Learning with few or imperfect data in the context of autonomous driving, focusing on overcoming challenges such as limited, noisy, imbalanced datasets by developing one/few-shot learning algorithms, synthetic data generation/augmentation, and leveraging transfer learning techniques.

PAPERS

Static Segmentation by Tracking: A Frustratingly Label-Efficient Approach to Fine-Grained Segmentation — Zhenyang Feng, Zihe Wang, Saul Ibaven Bueno, Tomasz Frelek, et al. **ICML 2025** [Submitted]

Impact of Camera Defects on Detection in Autonomous Driving — Tomasz Frelek, Guanqun Song, Ting Zhu **SenSys 2025** [Submitted]

PROJECTS & COMPETITIONS

NASA Airport Throughput Prediction Challenge [\[link\]](#)

Fall 2024

Time Series Forecasting Challenge

- Collaborating on a small team to develop a regression model to forecast airport throughput for U.S. airports, predicting flight arrivals over 15-minute intervals up to 3 hours into the future using flight and weather data.
- Constructing a comprehensive data pipeline from scratch to automate the process of data extraction, feature engineering, model training, and prediction, ensuring scalability and reproducibility
- Processing and analyzing large datasets, including FUSER, METAR, TAF, and CWAM data, totaling over 200 GB, to extract actionable features for model training and prediction.
- Using deep-learning network architectures such as LSTM and machine-learning frameworks like LightGBM to develop and implement a time series analysis model capable of predicting airport throughput; achieved 78.77% accuracy (9th/53 teams)

CVPR 2025 Anti-UAV Challenge [\[link\]](#)

Spring 2025

CVPR 2025 Workshop Challenge

- Developed an anti-UAV (drone) tracking and detection system by adapting and training two state-of-the-art computer vision models: SiamFC (Siamese Fully-Convolutional Network) for object tracking and YOLOv11 for object detection.
- Engineered a custom data pipeline to subsample a large-scale dataset into a format suitable for both the SiamFC and YOLOv11 models.
- Implemented and evaluated the models, calculating metrics such as mean IoU, success rate, and mAP50 to validate performance.

CLUBS & LEADERSHIP

Buckeye AutoDrive

Columbus, OH

Perception Team

Dec 2024 – May 2025

- Collaborating with cross-functional teams to create an autonomous vehicle that can navigate urban driving courses as described by SAE Level 4 automation.
- Developing and integrating perception algorithms for real-time object detection, classification, and tracking using LIDAR point cloud data, live video recording, and auxiliary sensor support to enhance autonomous vehicle situational awareness.
- Designing and implementing deep learning models utilizing PyTorch for pedestrian detection and semantic segmentation, achieving 90% accuracy in trials.

O.S.U. AI Club

Columbus, OH

Treasurer

May 2023 – May 2024

- Managed over \$10,000 in club finances and sponsorships, created detailed budgets, tracked expenses, and ensured funds were strategically allocated to enhance club activities, resources, and outreach efforts.
- Organized HackAI, an AI-focused hackathon; Oversaw all aspects of event planning, including participant recruitment, logistics, securing sponsorships, and distributing \$2,000 in prize money; resulted in over 150 participants and showcased innovative AI projects.
- Developed and lead a machine learning project series, designed with educational modules using scikit-learn and OpenCV to teach members practical skills, culminating in a handwriting detection and translation project that integrated real-time image processing and OCR techniques.
- Organized a university wide AI research expo with 15+ faculty presenters and 100+ students to connect eager students with professors conducting exciting AI research